

Danil Merinov

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EDUCATION

Drexel University

Bachelor of Science in Computer Science

Bachelor of Science in Mechanical Engineering

Philadelphia, PA

September 2019 – June 2024

Cumulative GPA: 3.39

EXPERIENCE

Drexel University

Data Analyst

August 2024 – Present

Philadelphia, PA

- Manage and analyze sensor data from LiDAR, RADAR, cameras, and GPS, leveraging statistical modeling
- Perform quantitative analyses using R and Python, automating report generation to deliver insight to stakeholders
- Train and optimize an open-source YOLOv7 model for real-time object detection, improving accuracy by 20%

Macquarie Group

Software Engineer Intern

September 2023 – April 2024

Philadelphia, PA

- Designed, developed, and deployed the team's homepage using JavaScript, HTML, CSS, and Hugo via Bitbucket
- Developed a password rotator using Python, configured load balancer with AWS CloudFormation, wrote unit tests
- Automated Fortify and BlackDuck security code scans with Bash scripts, containerized the solution with Docker
- Migrated a Python ETL application from Pandas to Polars, achieving a 500% increase in processing speed

Meusic

Software Engineer Intern

December 2022 – July 2023

Philadelphia, PA

- Designed and developed application's authentication and user functionalities, created automated endpoint testing
- Built a real-time chat feature using Django Channels and Redis, enabling cross-platform communication
- Containerized and maintained the application using Docker, MySQL, and a reverse proxy, facilitating scalability
- Deployed the application on a DigitalOcean CentOS Linux server, ensuring seamless operation and compatibility

Drexel University

Undergraduate Researcher

June 2022 – September 2022

Philadelphia, PA

- Studied the impact of electromagnetic field on drone performance by developing and testing a modular transformer
- Utilized MATLAB and Python to analyze flight data, identifying key performance metrics and generating insights
- Designed AutoCAD drawings for drones and power line simulations, enhancing visualization capabilities

University of Pennsylvania

Undergraduate Researcher

March 2021 – November 2021

Philadelphia, PA

- Consolidated research on polymerizing fibrin gels by conducting experiments using rheometer, calibrating it daily
- Studied the viscoelastic properties of hydrogels by varying fibrinogen, thrombin, calcium and plasma levels
- Coauthored a published article: <https://www.sciencedirect.com/science/article/abs/pii/S1751616122002405>

PROJECTS

Ingesting 13G - Capstone Project with Bloomberg L.P.

- Developed Python scripts to efficiently retrieve, process, and store 13G documents into S3 and DynamoDB
- Leveraged AWS services such as API Gateway and Lambda to perform CRUD operations on document data
- Used AWS SageMaker for data labeling, trained a NER model for data point extraction with 98% success rate

InsightAI - AI-driven Health Insights

- Developed an iOS app using SwiftUI and Apple HealthKit to generate user health insights with Llama 2
- Utilized SpeziLLM for on-device query processing, managed file caching for efficient model retrieval and execution

TuneLink - Split Sheets Generation Software

- Designed and implemented an automated CI/CD pipeline using GitHub Actions, reducing deployment time
- Architected and developed RESTful APIs using Java, Spring Boot, and MongoDB for management of artists' songs

SKILLS

Languages: Python, Java, C, C++, Swift, SQL, JavaScript, TypeScript, HTML, CSS, Bash, MATLAB

Tools: React, Angular, Spring Boot, Django, Git, LaTeX, Docker, MySQL, PostgreSQL, MongoDB, Airflow, DBT